C. Realism of Baseline Assumptions

3. **The baseline assumptions are consistent with past economic performance and the current government's fiscal consolidation commitment.** Staff assumes some contractionary effects from fiscal consolidation and expects that Aruba's real GDP growth will pick up over the medium term to about 1.1 percent. Over the medium term, a gap remains between staff and the authorities' forecasts.

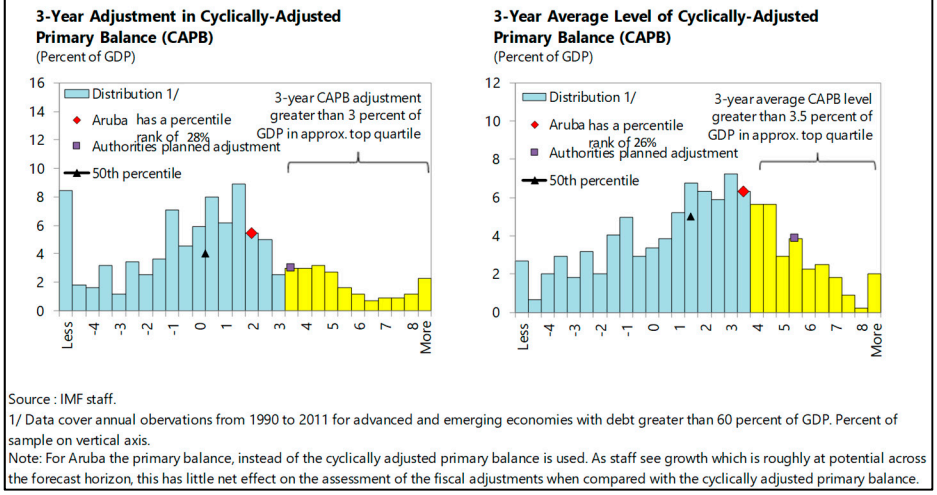
4. **Given Aruba's sizable debt burden and financing needs, the primary balance is expected to exceed its debt-stabilizing level of 1.7 percent of GDP over the projection period by a significant margin.** Under staff's baseline scenario, the primary balance should exceed 3.7 percent of GDP over 2019-24 to significantly lower the debt ratio. Enacting additional fiscal consolidation measures would further reduce debt sustainability concerns.

5. **Under the baseline scenario, the forecasted adjustments are large but within a normal historical range, as shown by the DSA's realism module.** Staff's projected maximum 3-year adjustment in the primary balance over the forecast period has a percentile rank of 28 among all advanced and emerging surveillance economies with debt greater than 60 percent of GDP (see chart below). The 3-year average level of projected primary balance is in the 26th percentile.

D. Realism of the Authorities' Planned Adjustment

6. **The authorities' fiscal targets are tighter than those of staff's baseline.**[3] Under the authorities' fiscal plan, the peak 3-year adjustment in the primary balance would be near the 20th percentile of realized outcomes, making this targeted adjustment slightly less likely to be realized. However, staff emphasizes that the macroeconomic environment of low oil prices, small multipliers, and robust near-term U.S. growth makes this an opportune time for Aruba to engage in fiscal reforms.

Assessing the Realism of Projected Fiscal Adjustment



Source : IMF staff.
1/ Data cover annual obervations from 1990 to 2011 for advanced and emerging economies with debt greater than 60 percent of GDP. Percent of sample on vertical axis.
Note: For Aruba the primary balance, instead of the cyclically adjusted primary balance is used. As staff see growth which is roughly at potential across the forecast horizon, this has little net effect on the assessment of the fiscal adjustments when compared with the cyclically adjusted primary balance.

[3] This active scenario is discussed in greater detail in paragraphs 18 and 19 of the main text.

©International Monetary Fund. Not for Redistribution

Figure 1. Aruba: Public Sector Debt Sustainability Analysis (DSA) - Baseline Scenario

(in percent of GDP unless otherwise indicated)

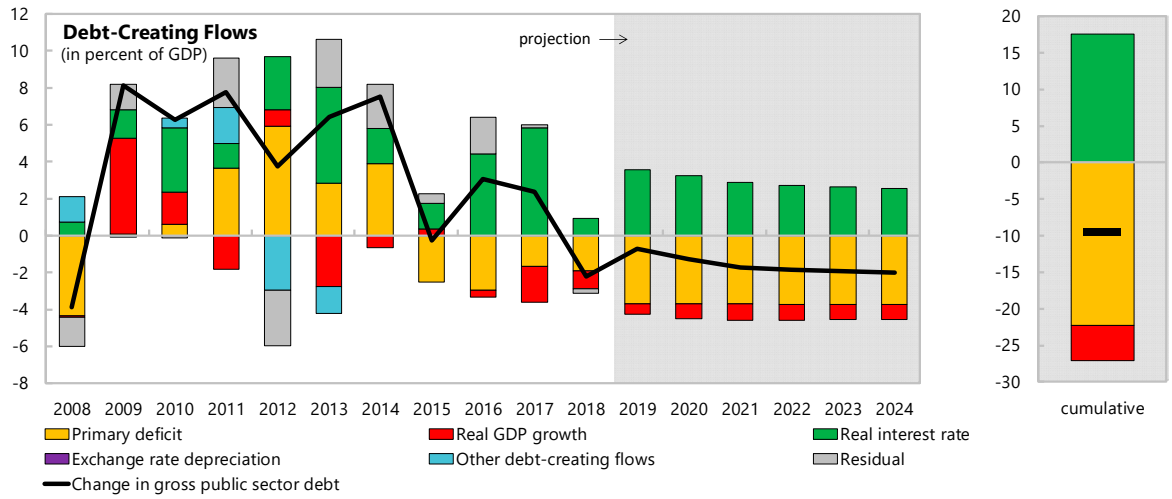
Debt, Economic and Market Indicators [1/]

	Actual			Projections					
	2008-2016 [2/]	2017	2018	2019	2020	2021	2022	2023	2024
Nominal gross public debt	66.7	86.7	84.5	83.8	82.5	80.8	79.0	77.0	75.0
Public gross financing needs	7.6	10.1	9.1	9.6	11.5	10.9	11.3	10.5	9.9
Real GDP growth (percent)	-0.8	2.3	1.2	0.7	1.0	1.1	1.1	1.1	1.1
Inflation (GDP deflator, percent)	1.1	-1.5	4.1	1.2	1.7	2.2	2.2	2.2	2.2
Nominal GDP growth (percent)	0.3	0.8	5.4	1.9	2.8	3.3	3.3	3.4	3.4
Effective interest rate (percent) [4/]	5.6	5.6	5.0	5.5	5.9	5.8	5.7	5.7	5.7

As of March 30, 2019		
Sovereign Spreads		
EMBIG (bp) 3/		300
5Y CDS (bp)		n.a.
Ratings	Foreign	Local
Moody's	n.a.	n.a.
S&Ps	BBB+	BBB+
Fitch	BBB-	BBB-

Contribution to Changes in Public Debt

	Actual			Projections							
	2008-2016	2017	2018	2019	2020	2021	2022	2023	2024	cumulative	debt-stabilizing primary balance [9/]
Change in gross public sector debt	4.3	2.4	-2.2	-0.7	-1.3	-1.7	-1.9	-1.9	-2.0	-9.5	
Identified debt-creating flows	3.5	2.2	-2.0	-0.7	-1.3	-1.7	-1.9	-1.9	-2.0	-9.5	
Primary deficit	0.8	-1.7	-1.9	-3.7	-3.7	-3.7	-3.7	-3.7	-3.7	-22.3	1.7
Primary (noninterest) revenue and grants	25.0	25.4	25.7	27.0	27.0	27.0	27.0	27.0	27.0	162.2	
Primary (noninterest) expenditure	25.8	23.7	23.8	23.3	23.3	23.3	23.3	23.3	23.3	139.9	
Automatic debt dynamics [5/]	2.8	3.9	-0.1	3.0	2.4	2.0	1.9	1.8	1.7	12.8	
Interest rate/growth differential [6/]	2.8	3.9	-0.1	3.0	2.4	2.0	1.9	1.8	1.7	12.8	
Of which: real interest rate	2.5	5.8	0.9	3.6	3.3	2.9	2.7	2.6	2.5	17.6	
Of which: real GDP growth	0.3	-2.0	-1.0	-0.6	-0.8	-0.9	-0.9	-0.8	-0.8	-4.8	
Exchange rate depreciation [7/]	0.0	0.0	0.0	...	...	...	...	...	...	...	
Other identified debt-creating flows	-0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
Privatization/Drawdown of deposits (+fina	-0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
Contingent liabilities	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
Please specify (2) (e.g., ESM and Euroarea l	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
Residual, including asset changes [8/]	0.8	0.1	-0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	



Source: IMF staff.

1/ Public sector is defined as central government.

2/ Based on available data.

3/ Long-term bond spread over U.S. bonds.

4/ Defined as interest payments divided by debt stock (excluding guarantees) at the end of previous year.

5/ Derived as [(r - π(1+g) - g + ae(1+r)]/(1+g+π+gπ)) times previous period debt ratio, with r = interest rate; π = growth rate of GDP deflator; g = real GDP growth rate; a = share of foreign-currency denominated debt; and e = nominal exchange rate depreciation (measured by increase in local currency value of U.S. dollar).

6/ The real interest rate contribution is derived from the numerator in footnote 5 as r - π (1+g) and the real growth contribution as -g.

7/ The exchange rate contribution is derived from the numerator in footnote 5 as ae(1+r).

8/ Includes asset changes and interest revenues (if any). For projections, includes exchange rate changes during the projection period.

9/ Assumes that key variables (real GDP growth, real interest rate, and other identified debt-creating flows) remain at the level of the last projection year.

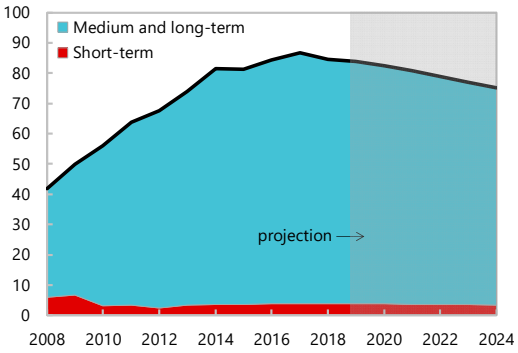
©International Monetary Fund. Not for Redistribution

Figure 2. Aruba: Public DSA - Composition of Public Debt and Alternative Scenarios

Composition of Public Debt

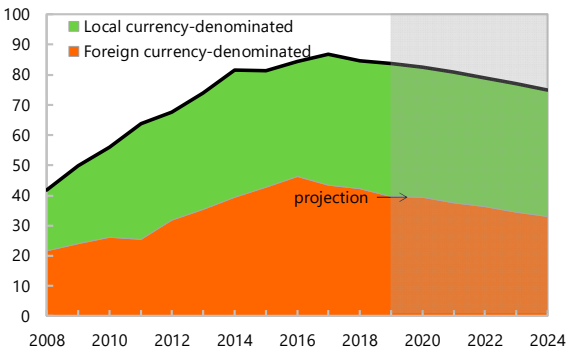
By Maturity

(percent of GDP)

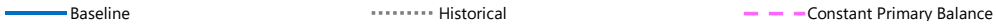


By Currency

(percent of GDP)

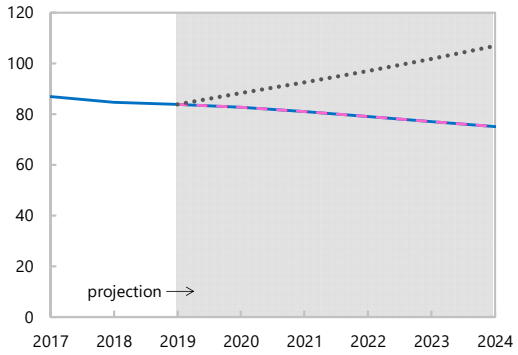


Alternative Scenarios



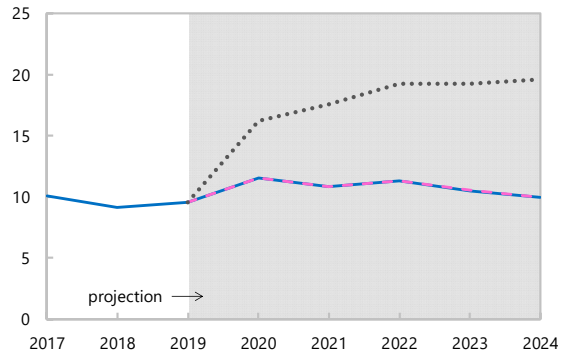
Gross Nominal Public Debt

(percent of GDP)



Public Gross Financing Needs

(percent of GDP)



Underlying Assumptions

(in percent)

Baseline Scenario	2019	2020	2021	2022	2023	2024
Real GDP growth	0.7	1.0	1.1	1.1	1.1	1.1
Inflation	1.2	1.7	2.2	2.2	2.2	2.2
Primary Balance	3.7	3.7	3.7	3.7	3.7	3.7
Effective interest rate	5.5	5.7	5.8	5.7	5.7	5.7
Constant Primary Balance Scenario						
Real GDP growth	0.7	1.0	1.1	1.1	1.1	1.1
Inflation	1.2	1.7	2.2	2.2	2.2	2.2
Primary Balance	3.7	3.7	3.7	3.7	3.7	3.7
Effective interest rate	5.5	5.7	5.8	5.7	5.7	5.7

Historical Scenario	2019	2020	2021	2022	2023	2024
Real GDP growth	0.7	-0.4	-0.4	-0.4	-0.4	-0.4
Inflation	1.2	1.7	2.2	2.2	2.2	2.2
Primary Balance	3.7	-0.8	-0.8	-0.8	-0.8	-0.8
Effective interest rate	5.5	5.7	5.8	5.9	6.0	6.0

Source: IMF staff.

©International Monetary Fund. Not for Redistribution